

## Lecture Note

- (A) Introduction
- (B) Covariance Stationary
- (C) Weakly Dependent Time Series
- (D) Autoregressive Process
- (E) Appendix

## (A) Introduction

- The heteroskedasticity and serial correlation problems violate the Gauss-Markov assumptions. The OLS estimators are no longer best among linear estimators. Standard errors computed using the regular OLS formula are not correct, and are generally underestimated.
- If a lagged dependent variable  $y_t$  is used as an explanatory variable, OLS estimators are mostly biased and inconsistent. Because, errors are usually correlated with the lagged  $y_t$ .
  - For instance,  $y_t = a_0 + a_1 y_{t-1} + u_t$ , where  $u_t = \rho u_{t-1} + e_t$
- The univariate time series models are a class of specifications where one attempts to *model* and/or best time series variable using information contained in their own past values. For example, we can model inflation rate, number visitor arrivals, electricity consumption, housing rental using their past values.
- A univariate model is useful when a structural model is inappropriate or unknown, where a structural model is a model trying to explain the changes in

a variable by reference to the movements in the current and the past values of other explanatory variables.

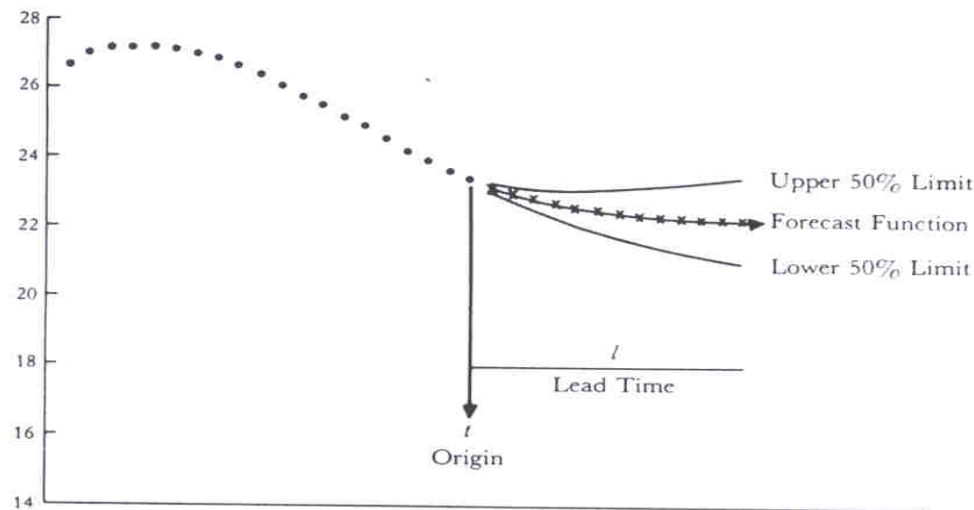
- This note introduces a few important fundamental concepts.

## (B) Covariance Stationary

- As discussed in Lecture Note 3, we need to make sure that the time series process is at least covariance stationary.
- Practically speaking, if we want to understand the relationship between two or more time series variables using regression analysis, we need to assume some sort of stable relationship. If we allow the relationship between  $y_t$  and  $x_t$  variables to change arbitrarily over time, then we cannot hope to learn much about a change in one variable affects the other variable.
- If the time series is not stationary, we can study its behavior only for the time period consideration. Each set of time series data will therefore be for a particular episode. In addition, for the purpose of forecasting time series may be of little practical value.

## Example: Hypothetical Time Series

Here, at time  $t$ , 20 historical values of a time series are available. The graph illustrates 13 forecast values from time  $t$  with the 50% probability limit. Simply put, the future values will fall within the band, with 50% of the probability. The band is widening when the time goes by.



Source: Time Series Analysis Forecasting Series Analysis (BJR 2008)

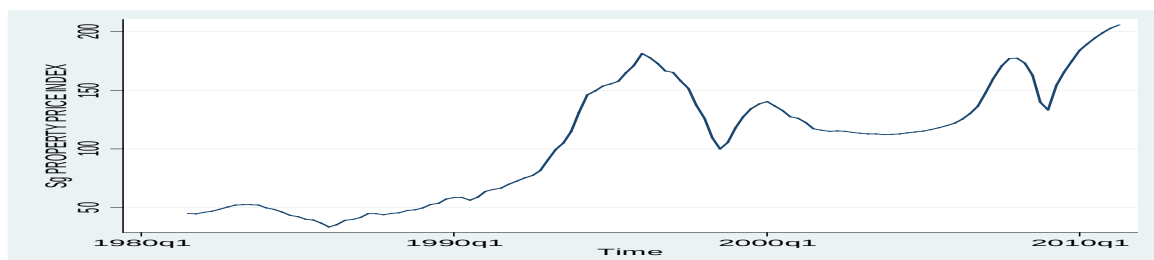
- **Definition:** A stochastic process  $\{y_t: t = 1, 2, \dots\}$  with finite second moment,  $E(y_t^2) < \infty$  is **covariance stationary** if
  - (i)  $E(y_t)$  is constant, and  $Var(y_t)$  is constant.
  - (ii) For any  $t$ ,  $h \geq 1$ ,  $Cov(y_t, y_{t-h})$  depends only on  $h$  (time distance) and not on  $t$  (specific time period).
- So, covariance stationary (known as weakly stationary) satisfies the following conditions.
  - (i)  $E(y_t) = \mu$ , and  $Var(y_t) = E(y_t - \mu)(y_t - \mu) = \sigma^2 < \infty$
  - (ii)  $Cov(y_t, y_{t-h}) = E(y_t - \mu)(y_{t-h} - \mu) = \gamma_h$ , where  $h \geq 1$ .  
If  $h = 0$ ,  $Cov(y_t, y_{t-0}) = Var(y_t) = \sigma^2$

Example: Are the following time series variables covariance stationary?

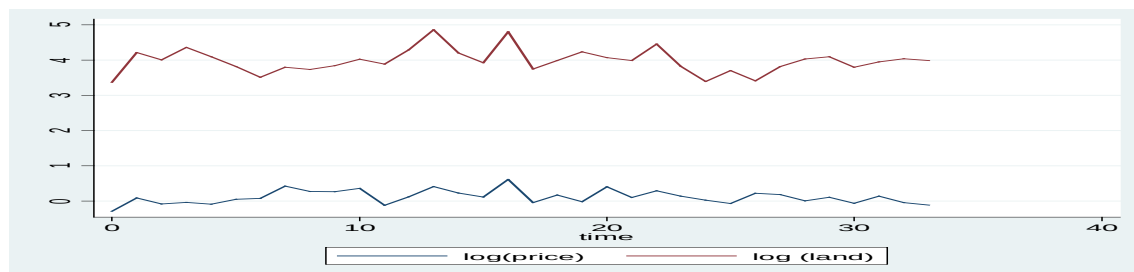
Singapore Inflation Rate



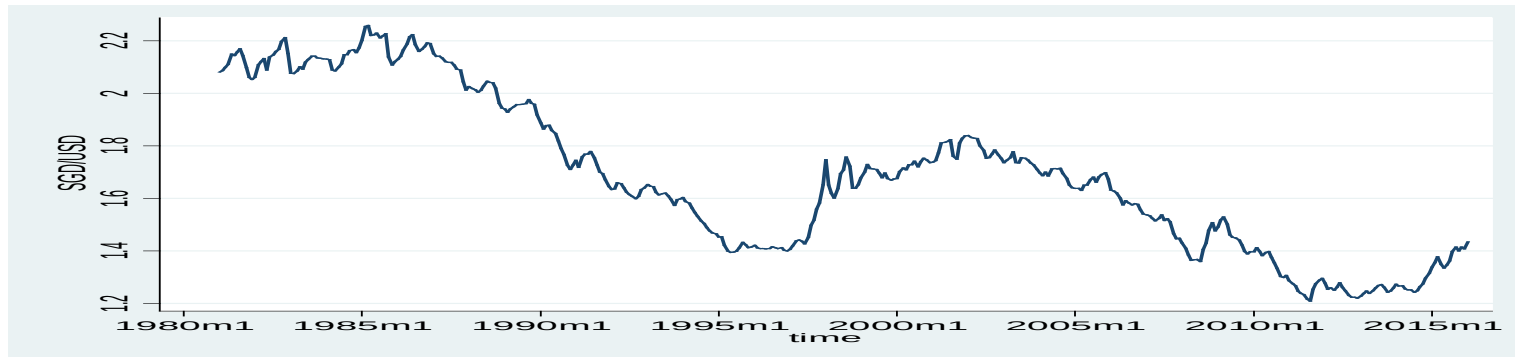
Singapore Housing Price Index



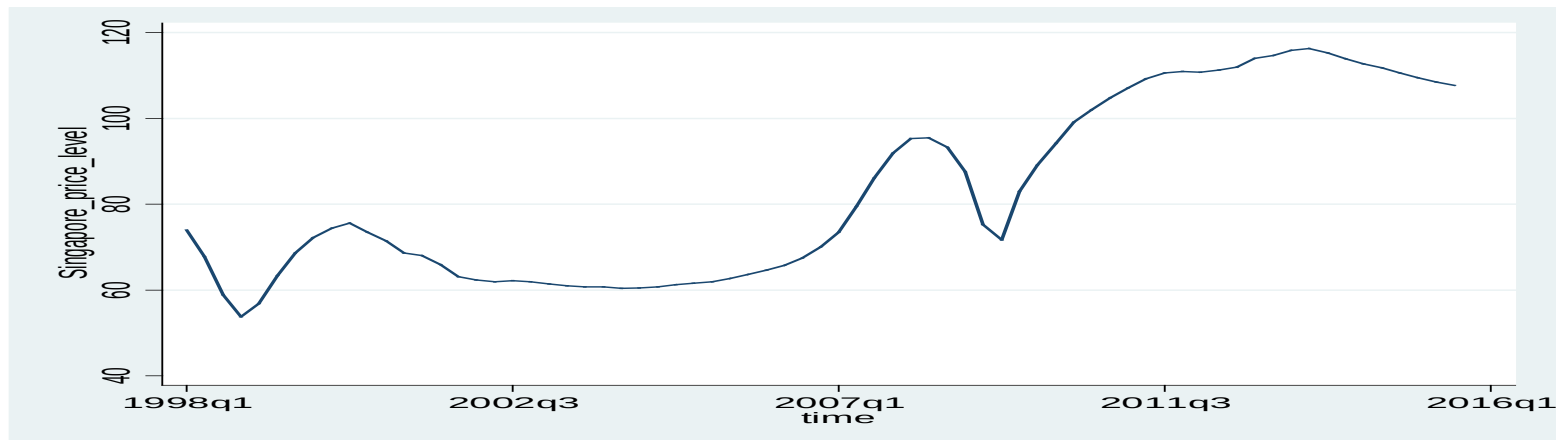
Sugarcane price and planted area in Bangladesh



Example: SGD/USD exchange rate (monthly data)



Example: Singapore Property Price Index (quarterly data)



- **Recap** (Principles of Statistics)

**Definition:** The **autocovariance**, e.g.,  $E(y_t - \mu)(y_{t-h} - \mu) = \gamma_h$ ,  $h \geq 1$ , is the covariance between  $y_t$  and  $y_{t-h}$ , where  $h \neq 0$ . The autocovariance determines how  $y_t$  is related to its own past (or future) values.

**Definition:** If we normalize the autocovariance ( $\gamma_h$ ) by the variance ( $\gamma_0$ ), we obtain the **autocorrelation function**.

$$\begin{aligned}
 ACF &= \text{Corr}(y_t, y_{t-h}) \\
 &= \frac{E(y_t - \mu)(y_{t-h} - \mu)}{\text{Var}(y_t)} \\
 &= \frac{E(y_t - \mu)(y_{t-h} - \mu)}{E(y_t - \mu)(y_t - \mu)} \\
 &= \frac{\gamma_h}{\gamma_0}
 \end{aligned}$$

If as  $h$  increases, the acf decreases very quickly, then it is weakly dependent

## (C) Weakly Dependent Time Series

- **Definition:** A covariance stationary time series is **weakly dependent** if the correlation between  $x_t$  and  $x_{t+h}$  goes to zero “sufficiently quickly” as  $h \rightarrow \infty$ . (Because of covariance stationarity, the correlation does not depend on the starting point,  $t$ .)
- **Definition:** Covariance stationary sequences where  $\text{Corr}(x_t, x_{t+h}) \rightarrow 0$  as  $h \rightarrow \infty$  are said to be **asymptotically uncorrelated**. Intuitively, this is how we will usually characterize weak dependence. As the variables get farther apart in time, the correlation between them becomes smaller and smaller.
- Weak dependence places restrictions on how strongly related the random variables  $x_t$  and  $x_{t+h}$  can be as the time distance between them ( $h$ ) increases.

Example: As the variables get farther apart in time, the covariance between them becomes smaller and smaller, and eventually it becomes 0.

$$\text{Cov}(\text{inf}_{2023, \text{Feb}}, \text{inf}_{2032, \text{Jan}}) = 0; \text{Cov}(\text{interest\_rate}_{1982}, \text{interest\_rate}_{2022}) = 0$$

- Weak dependence is important for regression analysis, because essentially, it replaces the assumption of random sampling in implying that the law of large numbers (LLN) and the central limit theorem (CLT) hold. So, stationary, weakly dependent time series are ideal for use in multiple regression analysis.

- **Recap** (Principles of Statistics)

The sample average  $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$ , where  $\bar{X}$  is a function of a random sample, is approximately normally distributed with a mean  $\mu$  and a variance  $\frac{\sigma^2}{n}$ . In addition,  $\bar{X}$  converges to  $\mu$  in probability.

Law of Large numbers:  $\bar{X} \approx \mu$ , when  $n \rightarrow \infty$ .

Central limit theorem:  $\bar{X} \sim N(\mu, \frac{\sigma^2}{n})$  or  $\sqrt{n} \left( \frac{\bar{X} - \mu}{\sigma} \right) \sim N(0, 1)$

- The simplest example of a weakly dependent time series is an independent, identically distributed sequence: a sequence that is independent is trivially weakly dependent.

Example: Moving average process of order 1,  $MA(1)$ , is a more interesting example of a weakly dependent sequence. Below is the  $MA(1)$ .

$$x_t = u_t + \alpha u_{t-1}, t = 1, 2, 3, \dots$$

where  $\{u_t: t = 0, 1, 2, 3, \dots\}$  is an i.i.d. sequence with  $E(u_t) = 0$  and  $Var(u_t) = \sigma^2$ .  $x_t$  is a weighted average of  $u_t$  and  $u_{t-1}$ ; and in the next period,  $x_{t+1}$  is a weighted average of  $u_{t+1}$  and  $u_t$ .

Obviously,

$$Cov(x_t, x_{t+1}) = Cov(u_t + \alpha u_{t-1}, u_{t+1} + \alpha u_t) = \alpha \sigma^2$$

$$Var(x_t) = (1 + \alpha)^2 \sigma^2$$

$$Corr(x_t, x_{t+1}) = \frac{Cov(x_t, x_{t+1})}{Var(x_t)} = \frac{\alpha}{(1 + \alpha)^2}$$

- In the previous example, due to the identical distribution assumption on the  $u_t$ ,  $\{x_t\}$  is clearly covariance stationary. It is also true that for  $MA(1)$ ,  $Cov(x_t, x_{t+h}) = 0$ , for any  $h \geq 2$ . Therefore,  $Corr(x_t, x_{t+h}) = 0$ .
- An  $MA(1)$  is a stationary, weakly dependent sequence, and the law of large numbers and the central limit theorem can be applied to  $\{x_t\}$ .

Example: Consider  $x_t = u_t + 0.8u_{t-1}$  where  $\{u_t: t = 0, 1, 2, 3, \dots\}$  is an i.i.d. sequence with zero mean and  $\sigma^2$  variance. Calculate  $E(x_t)$ ,  $Var(x_t)$ ,  $Cov(x_t, x_{t+1})$ ,  $Cov(x_t, x_{t-1})$ , and  $Corr(x_t, x_{t+1})$ .

$$E(x_t) = E(u_t) + 0.8E(u_{t-1}) = 0$$

$$Var(x_t) = Var(u_t) + 0.8^2 Var(u_{t-1}) = 1.64\sigma^2$$

$$Cov(x_t, x_{t+1}) = Cov(x_t, x_{t-1}) = 0.4\sigma^2$$

$$Corr(x_t, x_{t+1}) = \frac{0.4}{1.64} = 0.243$$

- **Discussion:**

(a) In the context of time series regressions with stationary, weakly dependent data: the variances of both the errors and the dependent variable do not change over time. By the law of large numbers, *R-squared* and adjusted *R-squared* both consistently estimate the population *R-squared*.

(b) Most of the standard statistical tests (e.g., F test and test) and estimation techniques (e.g., OLS and MLE) we used for the cross-sectional models are still valid for weakly dependent time series variables.

(c) Weak dependence also simplifies our statistical analysis task. Time series variables can be modeled by a simpler model that can still produce relatively accurate results.

## (D) Autoregressive Process

- **Definition:** The following  $\{y_t\}$  process is called an **autoregressive process of order one,  $AR(1)$** .

$$y_t = \rho y_{t-1} + u_t, \quad t = 1, 2, 3, \dots$$

The starting point in the sequence is  $y_0$  (at  $t = 0$ ), and  $\{u_t: t = 1, 2, 3, \dots\}$  is an i.i.d. sequence with  $E(u_t) = 0$  and  $Var(u_t) = \sigma^2$ . We also assume that the  $u_t$  are independent of  $y_0$  and that  $E(y_t) = 0$ .

- The crucial assumption for weak dependence of an  $AR(1)$  process is the stability condition:  $|\rho| < 1$ . We say that  $\{y_t\}$  is a **stable  $AR(1)$  process**.
- Due to the nature of  $u_t$ ,  $\{y_t\}$  is covariance stationary. In this case,  $E(y_t) = 0$ ,  $Var(y_t) = \frac{\sigma^2}{1-\rho^2}$ , and  $Cov(y_t, y_{t-h}) = \underbrace{\rho^h \sigma^2}_{1-\rho^2}$  is a function of the time distance  $h$  only.
- When  $|\rho| < 1$ ,  $Corr(y_t, y_{t-h}) = \frac{\rho^h}{1-\rho^2} \rightarrow 0$  as  $h \rightarrow \infty$ .

Example: What will the rate of inflation – the percentage increase in overall prices—be next year?

- Investors rely on forecast of inflation when deciding how much to pay for bonds.
- Economists at central banks, like the U.S. Federal Reserve Banks, use inflation forecasts when they set monetary policy.

Example:

$$(a) \ln(CPI_t) = \beta_0 + \beta_1 \ln(CPI_{t-1}) + u_t,$$

$$(b) \Delta GDP_t = \beta_0 + \beta_1 \Delta GDP_{t-1} + u_t,$$

$$(c) \Delta Property\_price\_index_t = \beta_0 + \beta_1 \Delta Property\_price\_index_{t-1} + u_t.$$

All models here are  $AR(1)$ , because for each model the dependent variable at time  $t$  depends on its value from  $(t - 1)$  and a random error term  $e_t$ .

- The  $AR(p)$  process is a natural extension of the  $AR(1)$  process. There are more lags of  $y_t$  included in the  $AR(p)$  model.

$$\begin{aligned}y_t &= \rho_0 + \rho_1 y_{t-1} + \rho_2 y_{t-2} + \cdots + \rho_p y_{t-p} + u_t \\ &= \rho_0 + \sum_{i=1}^p \rho_i y_{t-i} + u_t,\end{aligned}$$

Example:  $AR(2)$  process

$$\Delta interest\_rate_t = \rho_0 + \rho_1 \Delta interest\_rate_{t-1} + \rho_2 \Delta interest\_rate_{t-2} + u_t$$

For the sake simplicity, there are no other explanatory variables in the  $AR(2)$  model. The value  $y_t$  depends only on a history of its past values  $y_{t-1}$  and  $y_{t-2}$ .

The  $AR(2)$  model relies on correlations between values of the variable over time to help produce a forecast.

## (E) Appendix

[E.1] [Theory part **Not Tested, Advanced Econometrics/ Time Series Analysis**]

**Strictly Stationary Process:** The stochastic process  $\{y_t: t = 1, 2, \dots\}$  is strictly stationary if for every collection of time indices  $1 \leq t_1 \leq t_2 \leq \dots \leq t_m$ , the joint distribution of  $(y_{t_1}, y_{t_2}, \dots, y_{t_m})$  is same as the joint distribution of  $(y_{t_1+h}, y_{t_2+h}, \dots, y_{t_m+h})$  for all integers  $h \geq 1$ .

- Statistically speaking,  $F_{y_{t_1}, y_{t_2}, \dots, y_{t_m}}(y_1, \dots, y_m) = F_{y_{t_1+h}, y_{t_2+h}, \dots, y_{t_m+h}}(y_1, \dots, y_m)$
- Simply put, a stationary “time series” process is one whose probability distributions are stable over time in the following sense: If we take any collection of random variables in the sequence and then shift that sequence ahead  $h$  time periods, the joint probability distribution must remain unchanged.

Example:

- (a) The process also means the (joint) distribution of  $(y_1, y_2)$  is same as  $(y_{1+h}, y_{2+h})$ , where  $h \geq 1$ .
  - (b) The (joint) distribution of  $(y_1, y_4, y_8)$  is same as  $(y_{1+h}, y_{4+h}, y_{8+h})$  where  $h \geq 1$
- The strictly stationary process is often too strong for the reality. In addition, it can be very difficult to determine whether the data are generated (coming) from the strictly stationary process. Because, we only have one realization. Therefore, a weaker form of strictly stationary, i.e., covariance stationary, is enough for us to do econometric analysis.

[E.2] [Theory part **Not Tested, Advanced Econometrics/ Time Series Analysis**]

- Consider the  $AR(p)$  process  $y_t = \rho_0 + \sum_{i=1}^p \rho_i y_{t-i} + u_t$ . If all the eigenvalues are inside the unit circle, the  $AR(p)$  process is a convergent sequence.

$$y_t = \rho_0 + \rho_1 y_{t-1} + \cdots + \rho_p y_{t-p} + u_t$$

$$\Rightarrow y_t - \rho_1 y_{t-1} - \cdots - \rho_p y_{t-p} = \rho_0 + u_t$$

$$\Rightarrow (1 - \rho_1 L - \cdots - \rho_p L^p) y_t = \rho_0 + u_t, \text{ where } L \text{ is the lag operator.}$$

$$\Rightarrow y_t = \frac{\rho_0}{1 - \sum_{i=1}^p \rho_i L^i} + \frac{u_t}{1 - \sum_{i=1}^p \rho_i L^i}$$

- That means the  $y_t$  is stationary so long as the roots of  $1 - \sum_{i=1}^p \rho_i L^i$  are outside the unit circle. Note that the roots are for the “inverse characteristics equation”. Equivalently speaking, the eigenvalues of the  $y_t = \rho_0 + \sum_{i=1}^p \rho_i y_{t-i} + u_t$  are all inside the unit circle.
- More generally, for any  $p$ -th order difference equations. The stability conditions are as follows.
  - (a) Necessary Condition:  $\sum_{i=1}^p \rho_i < 1$  and
  - (b) Sufficient Condition:  $\sum_{i=1}^p |\rho_i| < 1$

### [E.3]

- An AR(1) process can be mathematically expressed as:  $y_t = \rho_0 \sum_{i=0}^{t-1} \rho^i + \rho^t y_0 + \sum_{i=0}^{t-1} \rho^i u_{t-i}$
- Taking the expectation both sides, the  $u$  terms vanish, as  $E(u_t) = 0$ .

$$E(y_t) = \rho_0 \sum_{i=0}^{t-1} \rho^i + \rho^t y_0$$

We can update  $t$  by  $h$  more periods,

$$E(y_{t+h}) = \rho_0 \sum_{i=0}^{t+h-1} \rho^i + \rho^{t+h} y_0$$

- For a finite series,  $E(y_t)$  is not equal to  $E(y_{t+h})$ . As a result, the sequence of  $\{y_t\}$  cannot be stationary. However, if  $t$  is very large, things change. Considering the extreme case that  $t \rightarrow \infty$ , we can calculate the limiting value of  $y_t$ .

$$\lim (y_t) = \rho_0 \sum_{i=0}^{\infty} \rho^i + \rho^{\infty} y_0 + \sum_{i=0}^{\infty} \rho^i u_{t-i}$$

Now, we take expectation on the equation,  $\lim (y_t) = \frac{\rho_0}{1-\rho} + \sum_{i=0}^{\infty} \rho^i u_{t-i}$ , then only  $\frac{\rho_0}{1-\rho}$  is left over. Thus, for a sufficiently long enough time periods, the mean of  $y_t$  is finite and time independent, i.e.,  $\mu = E(y_t) = \frac{\rho_0}{1-\rho}, \forall t$ .

- To calculate covariance, refer to the following.

$$\begin{aligned}
 & E[(y_t - \mu)(y_{t-h} - \mu)] \\
 &= E\{[u_t + \rho u_{t-1} + \rho^2 u_{t-2} + \dots][u_{t-h} + \rho u_{t-h-1} + \rho^2 u_{t-h-2} + \dots]\} \\
 &= \sigma^2 \times \rho^h (1 + \rho^2 + \rho^4 \dots) = \frac{\sigma^2 \rho^h}{1 - \rho^2}
 \end{aligned}$$

As we can see here, the covariance doesn't depend on the time  $t$ , it depends only on the length of the two time periods,  $h$